

TryHackMe - Linux Forensics

OS and account information

OS release information

To find the OS release information, we can use the `cat` utility to read the file located at `/etc/os-release`

User accounts

The `/etc/passwd` file contains information about the user accounts that exist on a Linux system. The output contains 7 colon-separated fields, describing username, password information, user id (uid), group id (gid), description, home directory information, and the default shell that executes when the user logs in

Group Information

The `/etc/group` file contains information about the different user groups present on the host. It can be read using the `cat` utility.

```
cat /etc/group
```

Sudoers List

A Linux host allows only those users to elevate privileges to `sudo`, which are present in the Sudoers list. This list is stored in the file `/etc/sudoers` and can be read using the `cat` utility. You will need to elevate privileges to access this file.

```
cat /etc/sudoers
```

Login information

In the `/var/log` directory, we can find log files of all kinds including `wtmp` and `btmp`. The `btmp` file saves information about failed logins, while the `wtmp` keeps historical data of logins. These files are not regular text files that can be read using `cat`, `less` or `vim`; instead, they are binary files, which have to be read using the `last` utility.

Authentication logs

Every user that authenticates on a Linux host is logged in the auth log. The auth log is a file placed in the location `/var/log/auth.log`

```
cat /var/log/auth.log
```

System Configuration

Hostname

The hostname is stored in the `/etc/hostname` file on a Linux Host. It can be accessed using the `cat` utility.

```
cat /etc/hostname
```

Timezone

Timezone information is a significant piece of information that gives an indicator of the general location of the device or the time window it might be used in. Timezone information can be found at the location `/etc/timezone` and it can be read using the `cat` utility.

```
cat /etc/timezone
```

Network Configuration

To find information about the network interfaces, we

can `cat` the `/etc/network/interfaces` file. The output on your machine might be different from the one shown here, depending on your configuration.

Similarly, to find information about the MAC and IP addresses of the different interfaces, we can use the `ip` utility.

```
ip address show
```

Active network connections

On a live system, knowing the active network connections provides additional context to the investigation. We can use the `netstat` utility to find active network connections on a Linux host

```
netstat -natp
```

Running processes

If performing forensics on a live system, it is helpful to check the running processes.

The `ps` utility shows details about the running processes.

```
ps aux
```

DNS information

The file `/etc/hosts` contains the configuration for the DNS name assignment. We can use the `cat` utility to read the hosts file.

```
cat /etc/hosts
```

The information about DNS servers that a Linux host talks to for DNS resolution is stored in the `resolv.conf` file. Its location is - `/etc/resolv.conf`

Persistence Mechanisms

Cron jobs

Cron jobs are commands that run periodically after a set amount of time. A Linux host maintains a list of Cron jobs in a file located at `/etc/crontab`

```
cat /etc/crontab
```

Service startup

Like Windows, services can be set up in Linux that will start and run in the background after every system boot. A list of services can be found in the `/etc/init.d` directory. We can check the contents of the directory by using the `ls` utility.

```
ls /etc/init.d/
```

.Bashrc

When a bash shell is spawned, it runs the commands stored in the `.bashrc` file. This file can be considered as a startup list of actions to be performed. Hence it can prove to be a good place to look for persistence.

```
cat ~/.bashrc
```

System-wide settings are stored in `/etc/bash.bashrc` and `/etc/profile` files, so it is often a good idea to take a look at these files as well.

Evidence of Execution

Sudo execution history

All the commands that are run on a Linux host using `sudo` are stored in the auth log. We can use the `grep` utility to filter out only the required information from the auth log.

```
cat /var/log/auth.log* |grep -i COMMAND|tail
```

Bash history

Any commands other than the ones run using `sudo` are stored in the bash history. Every user's bash history is stored separately in that user's home folder. Therefore, when examining bash history, we need to get the `bash_history` file from each user's home directory. It is important to examine the bash history from the root user as well, to make note of all the commands run using the root user as well.

```
cat ~/.bash_history
```

Files accessed using vim

The `Vim` text editor stores logs for opened files in `Vim` in the file named `.viminfo` in the home directory. This file contains command line history, search string history, etc. for the opened files. We can use the `cat` utility to open `.viminfo`.

```
cat ~/.viminfo
```

Log files

Syslog

The Syslog contains messages that are recorded by the host about system activity. The detail which is recorded in these messages is configurable through the logging level. We can use the `cat` utility to view the Syslog, which can be found in the file `/var/log/syslog`. Since the Syslog is a huge file, it is easier to use `tail`, `head`, `more` or `less` utilities to help make it more readable.

```
cat /var/log/syslog* | head
```

Auth logs

We have already discussed the auth logs in the previous tasks. The auth logs contain information about users and authentication-related logs.

```
cat /var/log/auth.log* |head
```

Third-party logs

Similar to the syslog and authentication logs, the `/var/log/` directory contains logs for third-party applications such as webserver, database, or file share server logs. We can investigate these by looking at the `/var/log/` directory.

```
ls /var/log
```